

School of AIR

Internal Curriculum for the AIR Founding Team

Read in order. Each module builds on the last. Everything is explained simply — no jargon without a definition.

Total reading time: ~2 hours

#	Module	What You'll Learn
01	The Problem	Why AI agents need identity
02	What AIR Is	What we built, how it works
03	How Trust Scores Work	The 5-component model
04	The Standards World	W3C, IETF, DIF, NIST
05	Building Blocks: DIDs and VCs	The W3C standards AIR uses
06	The Content Provenance Story	C2PA, CAWG, Adobe connection
07	Where AIR Fits	Our role in the ecosystem
08	What We Have vs. What We Claim	Honest assessment
09	Talking to People	Cheat sheets for every audience

Module 01: The Problem

The One-Sentence Version

AI agents are doing things in the world — writing code, managing data, creating content, making decisions — and nobody can verify who they are or whether they should be trusted.

Think of It Like This

Imagine you hire a contractor to renovate your kitchen. Before you let them into your house, you'd check:

- **Who are they?** (Name, company, license number)
- **Are they legit?** (Licensed, insured, references)
- **Can they do the work?** (Portfolio, certifications, reviews)
- **Should you trust them?** (Track record, complaints, ratings)

Now imagine there are thousands of contractors showing up at your door, and NONE of them have ID. No licenses. No ratings. No way to verify anything. That's the AI agent world right now.

What's an AI Agent?

An AI agent is software that acts on your behalf. Not just a chatbot that answers questions — an agent that actually DOES things:

- A code review agent that examines your pull requests
- A customer service agent that handles support tickets
- A financial analysis agent that processes your data
- A content creation agent that writes blog posts for you

These agents are getting more autonomous. They're making decisions. They're interacting with other agents. They're handling sensitive data.

The Three Problems

Problem 1: No Identity

When an AI agent contacts your system, there's no standard way to answer "who is this?"

- Is this agent really from the company it claims?
- Is it the same agent that did good work last month, or a new one impersonating it?
- Who built this agent? Who's responsible if it breaks something?

Today, the answer is: nobody knows.

Problem 2: No Trust

Even if you know who an agent is, how do you know whether to trust it?

- Has it been tested? Audited? Certified?
- Does it have a track record of reliable behavior?
- Is its source code available for review?
- Has anyone else vouched for it?

Today, you either trust everything or trust nothing. There's no middle ground.

Problem 3: No Portability

An agent has a great reputation on Platform A. You want to use it on Platform B. But:

- Platform A's identity system doesn't work on Platform B
- The agent's reputation doesn't transfer
- It has to start from zero on every new platform

This is like having a credit score that only works at one bank.

Why This Is Getting Urgent

Three trends are converging:

- 1. Agents are multiplying.** Every AI company is building agent capabilities. OpenAI, Anthropic, Google — plus thousands of startups.
- 2. Agents are talking to each other.** Anthropic's MCP and Google's A2A protocol are building the plumbing for agent-to-agent communication. But when Agent A talks to Agent B, neither can verify who the other is.
- 3. Regulators are watching.** NIST is actively working on AI agent identity. The EU AI Act will require traceability. Governments want to know: when an AI agent does something, who's accountable?

What's Missing

There's no neutral, independent registry where an agent can get a verifiable identity, earn a transparent trust score, present credentials that work across platforms, and build a portable reputation over time.

That's what AIR is building.

Key Takeaway

The problem isn't technical — it's infrastructural. The internet needed DNS (a neutral registry for domain names) before websites could work. The agent ecosystem needs something similar before agents can interoperate trustfully. AIR is that registry.

Module 02: What AIR Is

The Elevator Pitch (2 minutes)

"AIR is a neutral, open registry that gives AI agents verifiable identity and transparent trust scores. Think of it as the passport office plus credit bureau for AI agents. Any agent, from any platform, can register, get a unique ID, and earn a trust score based on objective criteria. It's built on open standards so it works everywhere — no vendor lock-in."

The Analogy

Real World	AIR Equivalent
Passport	AIR ID (unique identifier)
Credit Score	Trust Score (0-1000)
Business License	Verification Level
Background Check	Trust Score Components
Passport Office	AIR Registry

What We Actually Built (Today, Live, Working)

1. The Registry API

A real, working REST API at: agentidentityregistry.org/api/v1/

What it does:

- **Register an agent** — send name, creator info, capabilities, get back a unique AIR ID
- **Look up an agent** — send an AIR ID, get back the full identity + trust score
- **List all agents** — see every agent in the registry
- **Check if a name exists** — warns about duplicate names
- **Update an agent** — change description, capabilities, etc.
- **Delete an agent** — admin-only, soft delete (data preserved)

Runs on Cloudflare Workers (free tier), database on Cloudflare D1 (also free).

2. The AIR ID

Every registered agent gets a unique identifier: **AIR-7F3K-M9JQ-X2PL**

Format: AIR-XXXX-XXXX-XXXX

Generated by hashing the agent's identity info with SHA-256, encoding in base32, and adding a CRC32 checksum. The result is unique, verifiable, and human-readable.

3. The Trust Score

Every agent gets a score from 0 to 1000:

Component	Weight	What It Measures
Provenance	25%	Who made this agent?
Behavioral	25%	Track record (starts at 500)
Transparency	20%	Open source? Documentation?
Security	15%	Certifications?
Peer Attestations	15%	Third-party endorsements (starts at 300)

Grades: AAA (950+), AA (850+), A (700+), BBB (600+), BB (500+), B (400+), C (below 400)

4. The Website — agentidentityregistry.org

- **Homepage** (/) — what AIR is, how it works
- **Register** (/register) — form with live trust score preview
- **Lookup** (/lookup) — search any agent, see the full "identity passport"
- **About** (/about) — mission, values, timeline
- **Governance** (/governance) — how decisions are made
- **Blog** (/blog) — updates and announcements
- **Admin** (/admin) — protected dashboard

5. The GitHub Repository

github.com/ahnkwangwook-oss/agent-identity-registry

Full API source code, specification v0.1, trust methodology, governance docs, website source. Apache 2.0 licensed. GitHub Discussions enabled.

Key Numbers (April 2026)

- 8 demo agents (flagged as demo), 0 real agents
- 9 API endpoints live
- 3 standards bodies contacted (W3C, IETF, DIF)
- 1 NIST public comment submitted
- 1 standards body response received (DIF — Eric Scouten at Adobe)
- \$0 monthly infrastructure cost

Module 03: How Trust Scores Work

The Simple Version

A trust score is like a credit score for AI agents. Instead of measuring financial responsibility, it measures five things that tell you whether to trust an agent.

The Five Components

1. Provenance (25%) — "Where did you come from?"

About the agent's origins — who created it, are they verifiable?

Today's scoring: Creator DID +100, Creator name +100, Organization +100. Base 300, cap 600.

2. Behavioral (25%) — "What's your track record?"

How the agent actually performs over time. Starts at 500 (neutral, no history). Will change as behavioral data comes in.

3. Transparency (20%) — "Can we see how you work?"

Openness — is the code available, is there documentation?

Today's scoring: Open source +150, Code repo +100, Docs URL +100. Base 300, cap 650.

4. Security (15%) — "How well are you protected?"

Security posture — certifications, audit history.

Today's scoring: Each certification +100 (up to 3). Base 300, cap 600.

5. Peer Attestations (15%) — "Who vouches for you?"

Third-party endorsements. Starts at 300 (none yet).

The Formula

Total = (Provenance x 0.25) + (Behavioral x 0.25) + (Transparency x 0.20) + (Security x 0.15) + (Peer Attestations x 0.15)

Example — Well-documented open-source agent from an organization:

Provenance 600, Behavioral 500, Transparency 650, Security 500, Peer 300

= 150 + 125 + 130 + 75 + 45 = **525 (BB grade)**

Example — Anonymous agent with nothing verified:

Provenance 400, Behavioral 500, Transparency 300, Security 300, Peer 300

= 100 + 125 + 60 + 45 + 45 = **375 (C grade)**

The Live Score Preview

On the registration form (/register), a sidebar shows your estimated trust score in real time as you fill in fields. Every checkbox you tick, every URL you add, the score goes up. This encourages completeness.

What the Grades Mean

Grade	Think of it as...
AAA	"Would trust with my bank account"
AA	"Would deploy in production"
A	"Good for general use"
BBB	"Fine with oversight"
BB	"Use cautiously"
B	"Testing only"
C	"Don't trust"

Not Built Yet

Behavioral scoring over time, peer attestation submission, score changes from verification upgrades, dispute resolution, score history. These come as the registry gets real usage.

Module 04: The Standards World

Why Standards Bodies Matter

Standards are agreements about how things should work so they're compatible. For AI agents, there are no widely adopted standards yet for identity and trust. Multiple organizations are working on pieces.

The Key Organizations

NIST (National Institute of Standards and Technology)

US government agency. Creates standards and frameworks. Working on AI agent identity through CAISI initiative.

Our status: Public comment submitted April 3, 2026.

W3C (World Wide Web Consortium)

Creates web standards (HTML, CSS, URLs). Created two standards we build on — DIDs and VCs. Has an AI Agent Protocol Community Group.

Our status: Introduction email sent.

IETF (Internet Engineering Task Force)

Creates core internet protocols. Has AIMS (AI agent authentication) and WIMSE working groups. Authentication = "prove you are who you say you are." AIR adds: "should we trust you?"

Our status: Introduction email sent.

DIF (Decentralized Identity Foundation)

Consortium building decentralized identity tech. Home to CAWG (Creator Assertions Working Group). Members include Microsoft, IBM, Mastercard.

Our status: Invited to join as Contributor. Eric Scouten (Adobe) personally reached out.

How They Relate

- NIST says "agents need identity"
- W3C defines identity formats (DIDs, VCs)
- IETF defines authentication protocols
- DIF builds real implementations
- AIR is a working implementation that uses all of the above

Where AIR Sits

AIR is NOT a standards body. We're an IMPLEMENTER — a working system that uses the standards these bodies create.

Analogy: W3C defines what a passport looks like. IETF defines how border control verifies it. DIF tests the process. AIR is the passport office that actually issues passports.

How Standards Bodies Work (Practically)

- They move slowly (2-5 years for a standard)
- They value implementers (we have a working system — this matters)
- Participation is mostly free
- Communication is email + calls

Module 05: Building Blocks — DIDs and VCs

DIDs (Decentralized Identifiers)

Simple Explanation

A DID is like an email address, but you OWN it (no company can take it away), it's VERIFIABLE (cryptographically provable), and it works EVERYWHERE.

What One Looks Like

did:web:yourdomain.com — identity tied to a domain you control

did:key:z6MkhaXgBZ... — identity is a cryptographic key pair

How It Relates to AIR

When someone registers an agent, they provide a creator DID. This says "The creator of this agent controls this domain/key, and you can verify that."

The Key Insight

Normal identifiers (usernames, email) are controlled by the platform that issued them. DIDs are controlled by YOU. That's "decentralized."

VCs (Verifiable Credentials)

Simple Explanation

A digital version of a real-world credential (like a driver's license or diploma) with three superpowers:

1. **Cryptographically signed** — can't be tampered with
2. **Machine-readable** — software can automatically check it
3. **Selective disclosure** — show just the parts you want

The Three Parties

ISSUER (AIR) issues credential to **HOLDER** (Agent) who presents to **VERIFIER** (Anyone)

How It Relates to AIR

In AIR's design, trust scores and verification results are meant to be issued as VCs. Anyone who receives the credential can verify it without calling AIR's API — the credential carries its own proof.

What's Built vs. Planned

- Built: Agent identity documents follow JSON-LD structure (VC-compatible)
- Not built yet: Formal VC issuance, DID resolution endpoint, selective disclosure

How They Work Together

1. Agent registers → gets AIR ID linked to a DID
2. AIR issues a VC: "Agent AIR-7F3K has trust score 942"
3. Agent presents VC to third party

4. Third party verifies the signature

5. Trust established — no call to AIR needed

The VC is self-contained proof. This is powerful because it's decentralized.

Module 06: The Content Provenance Story

The Problem: "Who Made This?"

You see an image online. Was it created by a human or AI? If AI, which agent? Who controls it? Has it been modified? Nobody can answer these questions today.

C2PA: The Industry's Answer

C2PA = Coalition for Content Provenance and Authenticity. A "nutrition label for digital content."

Steering committee: Adobe, Microsoft, Google, Meta, OpenAI, Sony, BBC, Amazon. When all these companies agree on something, that's as close to "industry standard" as it gets.

How it works: Creator makes content. C2PA-compatible tool signs it with metadata (who, what, when). Anyone can verify the signature later.

CAWG: Adding Identity to C2PA

CAWG = Creator Assertions Working Group (part of DIF)

C2PA tells you WHAT tool made the content. CAWG adds WHO — verified identity credentials bound to the content.

Co-chaired by **Eric Scouten** (Adobe Principal Scientist) and **Scott Perry** (Digital Governance Institute).

The AI Agent Gap (Why Eric Reached Out)

The critical insight:

When a human uses Photoshop, C2PA records "made by [human identity]" — works fine.

When an AI agent creates content, the identity is TRANSACTIONAL — it only exists during the session. Once the agent finishes, there's no lasting record of which agent participated.

Eric calls this the difference between **transactional identity** (OAuth tokens, MCP sessions — disappear when session ends) and **archival identity** (must persist for years — royalty disputes, copyright claims).

AIR provides archival identity. An agent registered in AIR has a permanent, resolvable identity. This is exactly what CAWG needs.

How AIR + CAWG Could Work Together

1. Agent registered in AIR → gets AIR ID, linked to DID, trust score calculated
2. Agent creates content → C2PA manifest created
3. CAWG identity assertion added → "Created by agent AIR-7F3K, trust score 942 (AAA)"
4. Years later → anyone looks up AIR-7F3K, sees full identity and provenance

AIR becomes the **identity provider** for AI agents in the content provenance ecosystem.

Why This Is Bigger Than Eric

Content provenance is becoming legally required (EU AI Act, US executive orders, platform policies). If C2PA becomes the standard, every AI agent creating content will need verifiable identity. That's millions of agents needing what AIR provides.

Module 07: Where AIR Fits

The Layer Cake

Think of agent trust as layers. Each answers a different question:

Layer 4: TRUST ASSESSMENT — "Should I trust this agent?" (AIR lives here)

Layer 3: IDENTITY VERIFICATION — "Is this agent who it claims?" (DIDs, VCs)

Layer 2: AUTHENTICATION — "Can this agent prove its identity right now?" (AIMS, OAuth)

Layer 1: COMMUNICATION — "Can this agent talk to me?" (MCP, A2A, HTTP)

Most of the industry is building Layers 1 and 2. CAWG is building Layer 3 for content. AIR is the only project building Layer 4 as a neutral, independent service.

Complementary, Not Competitive

Partner	They Provide	AIR Provides
W3C	Identity standards	Implementation using those standards
IETF	Authentication protocols	Trust layer on top of auth
CAWG	Content provenance framework	Agent identity provider
DIF	Community, spec review	Working implementation
Platforms	Agents that need identity	Neutral cross-platform identity

Analogy: A passport office doesn't compete with airlines, border control, or hotels. It provides a foundational service they all need.

Strategic Advantages

1. **We're early** — no one else has a working neutral agent identity registry
2. **We're neutral** — platforms adopt neutral standards over competitors'
3. **Built on open standards** — not asking anyone to learn something new
4. **Working system** — standards bodies value implementers
5. **Regulators want this** — when regulation mandates agent identity, someone provides it

Module 08: What We Have vs. What We Claim

The Rule

Never claim more than what exists. Credibility, once lost, is almost impossible to rebuild.

What Is Real

Claim	Verifiable?
Live registry API	Yes — anyone can call it
Agents can register and get AIR IDs	Yes — try it
Trust scores calculated from 5 components	Yes — source in GitHub
Spec is published	Yes — public repo
NIST public comment submitted	Yes
DIF/Adobe responded	Yes — email thread
\$0 infrastructure	Yes — Cloudflare free tier

What Is NOT Real — Don't Claim These

False Claim	Reality
"We're a 501(c)(3)"	Not filed yet
"We have a board"	2 people — you and Kenny
"We have SDKs"	No SDKs. API only.
"Third-party audits quarterly"	No audits done
"Verification tiers operational"	All agents are self-verified
"We issue W3C VCs"	VC-compatible structure but no formal issuance
"Real agents registered"	All 8 are demo data

The "Two Truths" Test

Before saying anything: (1) Is it factually true RIGHT NOW? (2) Could someone verify it in 5 minutes?

If both yes — safe to say. If either no — reframe or don't say it.

Good Framing

- "Working prototype exploring standards alignment"
- "Early-stage project with a live API and published specification"
- "Open-source project, contributor team of two, seeking collaboration"

What We Can Be Proud Of

- Built a working system from scratch in weeks
- Published a genuine specification
- Submitted a substantive NIST public comment
- Got a personal response from a Principal Scientist at Adobe
- Open-sourced everything on \$0 infrastructure
- Identified a real gap that industry leaders independently confirm

Module 09: Talking to People

The Core Pitch (30 Seconds)

"AIR is an open-source registry that gives AI agents verifiable identity and transparent trust scores. Like a passport office for AI agents — any agent, from any platform, can register and get a unique ID and trust score. Built on W3C open standards, neutral and independent."

Cheat Sheet: Standards Body Person

Opening: "We're an early-stage open-source project building agent identity on W3C DIDs and Verifiable Credentials. We have a working API and published specification, and we're here to learn."

If asked about maturity: "Early stage. Working prototype with a live API. Two-person team. Looking for feedback."

Don't say: Don't claim W3C/DIF endorsement. Don't say "production-ready."

Cheat Sheet: Eric Scouten / CAWG

Opening: "We've been thinking about the exact problem you described — current agent identity is transactional. AIR provides persistent identity that survives beyond any session."

Questions to ask:

1. "What would an identity provider need to implement for CAWG compatibility?"
2. "Is the v1.3 VC support draft the right integration point?"
3. "How do you envision delegation chains — human authorizes agent to create content?"
4. "Existing implementations we should study?"

Don't say: Don't claim we issue VCs yet. Don't promise a timeline. Don't oversell as "production-grade."

Cheat Sheet: Potential Funder

Opening: "AIR is building neutral identity infrastructure for AI agents — think DNS for agent identity. We're the only independent project in this space with a working system."

Key points: Adobe/DIF validates need. NIST engagement shows policy relevance. \$0 infrastructure. Open source.

Cheat Sheet: Developer

Opening: "Open-source agent identity. Cloudflare Workers + D1, plain JavaScript, no dependencies. One GitHub repo."

Key points: API source is ~500 lines. Apache 2.0. Contribution guide included.

Universal Rules

1. Lead with the problem, not the solution

2. Use analogies — "passport office for AI agents"
3. Be honest about stage
4. Ask questions more than you make claims
5. Don't apologize for being early
6. Say "I don't know" when you don't know
7. Always offer the demo — "try it at agentidentityregistry.org/register"